# Smart Krishi: A Verified, Rule-Based Integrated Decision-Support Platform for Indian Farmers

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## Abstract

Indian smallholder farmers make decisions on crop choice, irrigation, fertilizer use, costing and selling largely without verified, location-specific information. Existing digital tools address only one of these decisions at a time, and many depend on static or unverifiable data. This paper reviews 32 studies on crop recommendation, forecast-based irrigation, soil-test-based fertilizer advice, price forecasting, market access and digital advisory adoption, and uses them to motivate Smart Krishi, an integrated platform of nine modules built on a stateless Java 17 / Spring Boot 3.3.4 backend, MySQL persistence and a lightweight Bootstrap frontend. The platform provides rule-based crop recommendation from eight soil and climate parameters, real-time Open-Meteo weather and evapotranspiration-based irrigation advice, ICAR-standard fertilizer calculations converted to commercial bag quantities, profit, ROI and break-even analysis, crop calendars, short-term price trend prediction, and a Haversine-based nearby mandi locator with a failover data strategy. A strict zero-fabricated-data policy governs every module. Sixteen integration test cases covering security, API timeouts, agronomic safety limits and data failover executed with no failures. We discuss limitations and outline planned upgrades, including ARIMA-based forecasting and field validation.

*Keywords: digital agriculture, decision-support system, crop recommendation, smart irrigation, fertilizer optimisation, mandi price, Spring Boot, JWT.*

## 1. Introduction

Agriculture supports a large share of India's population, yet marginal and small farmers, who form the bulk of the farming community, operate on small plots with limited access to soil testing, extension advice and market information [29], [30], [32]. Decisions that determine income are taken early and are hard to reverse: which crop to sow, how much to irrigate and fertilize, and where and when to sell.

Digital advisory tools can help. Voice-based advice built trust and changed input choices among cotton farmers in Gujarat [27], and mobile advisories helped rice farmers in Odisha cope with weather shocks [28]. However, adoption is uneven for smallholders [29]-[31], and most tools cover a single decision. A farmer who wants to choose a crop, check the weather, calculate fertilizer, estimate profit and find a market must use several unconnected sources.

This paper presents Smart Krishi, an integrated platform that chains these decisions for the same farmer and plot, and that exposes only verified, rule-based outputs. The contributions are: (1) a structured review of 32 studies mapped to the problems the platform addresses; (2) the design and implementation of a nine-module platform with a secure, stateless architecture; (3) agronomically safe, actionable

outputs such as fertilizer bag counts and explicit irrigation decisions; (4) a failover design that keeps the system honest when government APIs are down; and (5) an honest account of limitations relative to the literature.

# 2. Literature Review

## 2.1 Crop recommendation

Machine-learning crop recommenders typically take nitrogen, phosphorus, potassium, pH, temperature, humidity and rainfall as inputs. Kathoon et al. use neural network and CNN models [1]; a Madras Agricultural Journal study trains seven classifiers on the same seven variables and exposes them through a simple web form [2]; other work applies Random Forest and related ensembles for Indian crops [3], [4]. These systems show that the seven-variable input set is the accepted standard, but they depend on training datasets whose provenance and regional validity are often unclear, and they report accuracy rather than agronomic safety.

## 2.2 Weather-based irrigation and advisories

Irrigation-Advisor uses weather forecasts to compute soil evaporation and crop transpiration without on-site sensors and achieved water savings of up to 13% compared with farmer practice (lettuce was an exception) [5]. Pelosi et al. forecast crop evapotranspiration with a median bias of 0.2 mm and RMSE of 0.75 mm on the first forecast day [6]. In India, agromet advisory adopters showed better yield and yield attributes than non-adopters [7], and farmers were satisfied with advisories although the credibility of forecasts needed improvement for sustained use [8]. Open-Meteo provides free hourly and daily reference evapotranspiration (ET0) forecasts without an API key for non-commercial use [R3], making forecast-driven irrigation feasible for a student-built, low-cost platform.

## 2.3 Soil-test-based fertilizer recommendation

Soil degradation affects an estimated 120 million hectares (37% of India's area) and site-specific nutrient management depends on soil testing [10]. The Soil Test Crop Response (STCR) approach underpins several decision-support tools [12], [13], and ICRISAT derived recommendations from 40,265 soil samples in Odisha [9]. Rule-based and regression calculators such as FertiCal-P convert soil-test N, P2O5, K2O and pH into NPK advice [11]. The Soil Health Card programme found that over 90% of surveyed farmers had never tested their soil before the scheme and 92% were not using balanced fertilizer [R1]. Awareness of the card was 82%, but only 66% understood the advice and only 48% followed the recommended rates [14]. Where advice was followed, urea use fell without yield or financial loss [15], and adopters used about 20 kg/ha less nitrogen and 19 kg/ha less phosphorus [16]. The gap is therefore not only the quality of recommendations but their actionability.

## 2.4 Profitability analysis

Standard cost concepts (A2, B2, C2), cost of production (C2 divided by yield), net return and benefit-cost ratio are the accepted way to assess crop profitability [17]. Farmers usually lack a simple tool that applies these concepts to their own plot before sowing.

## 2.5 Price forecasting and market access

AGMARKNET is the main public price source. ARIMA has been applied to wheat prices, which showed marked seasonality [18], and to cotton prices in Haryana, where forecasts help farmers choose between nearby markets [19]. ARFIMAX uses market arrivals as an exogenous variable for mustard [20]; comparisons of ARIMA, RNN, LSTM, VAR, Random Forest and XGBoost exist for onion, potato and tomato [21]; and an interpretable web-based forecaster serves small farmers using over a thousand markets [22]. On the access side, transportation and especially market information improve the prices farmers receive [23]. Studies of e-NAM report mixed outcomes: limited price impact [24] and difficulties for farmers [25], while selling through mandis gave local-crop producers 13 to 73% higher prices than private traders [26]. Knowing which mandi is near and what it pays is therefore valuable regardless of platform-level reform.

## 2.6 Digital advisory adoption

Precision and digital agriculture face fragmented holdings, weak data infrastructure and unequal access [29], [30]. Service-based adoption outperforms individual technology ownership for smallholders [31], and one study of a Chhattisgarh sample found no adoption of precision technologies at all [32]. Advisory evaluations show trust and input adoption [27] and benefits for disadvantaged groups [28], but not systematic yield gains [27]. These findings argue for tools that are free to use, mobile-friendly, simple and trustworthy.

## 2.7 Summary of reviewed studies

Table 1 lists all 32 studies with their principal finding and the Smart Krishi module each informs.

| # | Study | Year | Key contribution / finding | Module |
|---|---|---|---|---|
| 1 | Kathoon et al. | 2025 | Neural network / CNN crop recommender using N, P, K, pH, temperature, humidity and rainfall. | M2 |
| 2 | AI-based Smart Crop Recommendation System (Madras Agric. J.) | 2025 | Trains seven ML models on the same seven inputs; farmer enters values through a simple web form. | M2 |
| 3 | A Machine Learning-Enabled System for Crop Recommendation (Eng. Proc. 67) | n/a | ML crop recommendation built from weather, soil and NPK data; reviews earlier classifier and DSS work. | M2 |
| 4 | A Reliable ML Framework for Crop Recommendation in Precision Agriculture | n/a | Random Forest / XGBoost-style classifiers on NPK, pH, humidity, temperature and rainfall for Indian crops. | M2 |
| 5 | Miras-Ávalos et al. (Irrigation-Advisor) | 2019-20 | Forecast-driven irrigation DSS without on-site sensors; water savings up to 13% vs farmer practice (lettuce was an exception). | M3 |
| 6 | Pelosi et al. | n/a | Forecasts crop evapotranspiration ahead of time (median bias 0.2 mm, RMSE 0.75 mm on day 1) to support irrigation advisories. | M3 |
| 7 | Validation of medium-range weather forecasts (PAU) | n/a | Agromet advisory adopters showed better yield and yield attributes than non-adopters. | M3 |
| 8 | Agromet advisory adoption, semi-arid southern India (Front. Sustain. Food Syst.) | 2023 | Farmers satisfied with agromet advisories, but forecast credibility needs improvement for sustained use. | M3 |

| # | Study | Year | Key contribution / finding | Module |
|---|---|---|---|---|
| 9 | Chander et al. (ICRISAT) | 2020 | Soil-test-based fertilizer recommendations derived from 40,265 soil samples across 30 districts of Odisha. | M4 |
| 10 | User-centered design of a fertilizer recommendation system (MIT thesis) | n/a | Mobile fertilizer recommendation engine for smallholders under site-specific nutrient management; notes ~120 Mha (37%) of Indian land affected by soil degradation. | M4 |
| 11 | FertiCal-P Android DSS | n/a | Rule-based + regression NPK calculator from soil-test N, P2O5, K2O and pH. | M4 |
| 12 | STCR & Dhartimitra, cowpea, Karnataka (Madras Agric. J.) | 2025 | STCR-based recommendations using decision-support software outperformed conventional recommendations. | M4 |
| 13 | DSS for fertilizer recommendation: a case study (Dharwad) | 2014 | Targeted-yield DSS built on Soil Test Crop Response research for farmers' fertilizer decisions. | M4 |
| 14 | The Soil Health Card scheme in India: lessons learned | n/a | Awareness 82%, but only 66% understood and only 48% followed recommended fertilizer rates. | M4 |
| 15 | Impact of Soil Health Card on urea use, Haryana (IJEE) | n/a | Following card advice reduced urea use without lowering yields or causing financial loss. | M4 |
| 16 | Socio-economic impact of Soil Health Card, Andhra Pradesh (IJAgS) | n/a | Adopters used ~20 kg/ha less N and ~19 kg/ha less P, reducing excess fertilizer in paddy. | M4 |
| 17 | Profitability of coarse cereals production in India | n/a | Uses cost concepts A2/B2/C2, cost of production = C2 / yield, net return and benefit-cost ratio. | M5 |
| 18 | Cariappa et al. (wheat price forecasting) | 2020 | ARIMA on AGMARKNET monthly prices; strong seasonality; stresses need to disseminate forecasts to farmers. | M7 |
| 19 | Kumar et al. (cotton, Haryana) | 2024 | ARIMA(1,1,1) on AGMARKNET data; forecasts help farmers choose between nearby markets. | M7 |
| 20 | Paul et al. (mustard, ARFIMAX) | 2022 | Long-memory model using arrivals as exogenous variable on daily Bharatpur and Agra prices. | M7 |
| 21 | Suresha & Uppaladinni | 2021 | Compares ARIMA, RNN, LSTM, VAR, Random Forest and XGBoost for onion, potato, tomato prices. | M7 |
| 22 | Interpretable produce price forecasting for small/marginal farmers | n/a | Pulls Agmarknet data for 1000+ markets and shows interpretable forecasts in a mobile-friendly web app. | M7 |
| 23 | Negi, Birthal et al. (Food Policy) | n/a | Access to transport and market information improves price terms; the information effect is stronger. | M8 |
| 24 | Venkatesh et al. (changing structure of agri marketing, e-NAM) | n/a | Studies e-NAM trade patterns, benefits and constraints; impact on farmers' prices reported as limited. | M8 |
| 25 | Sai Shiva, Nair & Thajudeen (IIMK WP 491) | n/a | Finds e-NAM digitization created difficulties and did not secure remunerative prices. | M8 |
| 26 | Agricultural marketing channels and market prices (Econ. Analysis & Policy 81) | 2024 | Local crop producers got 13-73% higher prices through mandis than private traders; export crops 19% less. | M8 |
| 27 | Cole & Fernando (Economic Journal) | 2021 | Voice-based advisory built trust and input adoption among cotton farmers; no systematic yield gains found. | All |
| 28 | Harigaya & Zhu (Precision Development) | 2025 | Mobile advisory helped farmers cope with weather shocks; disadvantaged groups benefited substantially. | All |

| # | Study | Year | Key contribution / finding | Module |
|---|---|---|---|---|
| 29 | Shaheen et al. | 2020 | Reviews challenges and opportunities of precision farming in India with farmer interviews near Hyderabad. | All |
| 30 | Whither digital agriculture in India? | n/a | Review: fragmented holdings, weak data infrastructure and unequal digital access slow smallholder adoption. | All |
| 31 | Rajesh & Malarkodi | n/a | Finds integrative agronomy + economics evidence is scarce; service-based adoption beats individual ownership. | All |
| 32 | Sathyamala | n/a | Argues digital tools may not benefit marginal farmers; in a Chhattisgarh sample none had adopted precision agriculture. | All |

*Table 1. Reviewed studies (n/a = not captured; confirm on Google Scholar)*

# 3. Problem Statement and Research Gap

From the review, five problems recur: (i) soil and input decisions without usable soil information [10], [14]; (ii) weather data not converted into crop-specific action [5], [8]; (iii) limited market and price information [18], [23]; (iv) uneven adoption of digital tools by smallholders [29]-[32]; and (v) point solutions that do not connect the decision chain. Table 2 contrasts representative existing systems with the proposed platform.

| System | Crop | Irrigation | Fertilizer | Price | Mandi |
|---|---|---|---|---|---|
| Crop ML systems [1]-[4] | Yes (learned) | No | No | No | No |
| Irrigation-Advisor [5] | No | Yes | No | No | No |
| FertiCal-P / STCR DSS [11]-[13] | No | No | Yes | No | No |
| Price forecasting [18]-[22] | No | No | No | Yes | No |
| Smart Krishi (proposed) | Yes (rule-based) | Yes | Yes (bags) | Basic trend | Yes (GPS) |

*Table 2. Coverage of representative systems versus Smart Krishi*

The research gap is an integrated, transparent and safe decision-support platform that covers the whole pre-sowing to sale chain, stays usable on low-cost devices, and refuses to display unverified data.

# 4. Proposed Solution: Smart Krishi

## 4.1 Design objectives

- Integrated: one farmer profile with multiple plots feeding all modules.
- Verified: strictly rule-based outputs grounded in ICAR thresholds, with zero fabricated telemetry; if a live source fails, the system says so instead of inventing values.
- Actionable: results expressed as decisions and physical quantities (pause irrigation, number of fertilizer bags, break-even price).
- Accessible: no heavy frontend framework, responsive layout, public tools usable before registration.
- Performant: backend responses within 250 ms and page load under 1.5 s on a 4G connection (design targets).

## 4.2 System architecture

Smart Krishi follows a distributed client-server architecture. The client communicates with a stateless Spring Boot service tier through asynchronous JSON Fetch API calls. Table 3 summarises the layers.

| Layer | Description |
|---|---|
| Presentation | Static HTML5 / CSS3 / ES6+ / Bootstrap 5.3 pages (api.js, auth.js, dashboard.js); asynchronous JSON Fetch calls with JWT injected in the header |
| Security | Spring Security 6, stateless JWT filter chain, BCrypt hashing, role-based access (ROLE_FARMER, ROLE_ADMIN, public) |
| Service / business logic | Crop, weather, fertilizer, profit, calendar, price-trend and mandi services; external clients for Open-Meteo and data.gov.in AGMARKNET |
| Persistence | Spring Data JPA over MySQL 8.0+ (InnoDB); H2 in MySQL mode for development; six core tables with referential integrity |

*Table 3. Architectural layers*

## 4.3 User roles and security

Access is controlled by stateless JWT role-based access control implemented with Spring Security 6 and a custom token filter. A registered farmer (ROLE_FARMER) manages plots and views recommendations; a platform administrator (ROLE_ADMIN) audits users, publishes government welfare schemes and seeds mandi prices; an unauthenticated user can use public tools such as the profit calculator, price prediction and mandi locator. Passwords are hashed with BCrypt. Removing a user cascades to the user's farm plots without violating foreign-key constraints.

## 4.4 Data model

The relational schema has six core tables with enforced referential integrity: users (id, email, password, role), farmer_profiles (user_id, land_area, soil_type, irrigation_type), farms (farmer_profile_id, plot_number, area_acres), crop_recommendations (stored inputs N, P, K, pH, temperature and the prescribed crop), market_prices (crop_name, mandi, modal_price, trend) and advisories (title, category, target_crop, details). MySQL 8.0+ with InnoDB is used in production and an in-memory H2 database in MySQL mode in development.

## 4.5 Functional modules and algorithms

| Module | Input | Method | Related work |
|---|---|---|---|
| M1 Farmer management | Registration, login, BCrypt hashing, multi-plot farm profiles | JWT RBAC, cascading delete of plots | [27], [28] |
| M2 Crop recommendation | N, P, K, pH, rainfall, humidity, temperature (8 soil/climate parameters) | Rule-based matching against agronomic ranges | [1]-[4] |
| M3 Weather and irrigation | GPS or city name -> Open-Meteo forecast and evapotranspiration | Geocode then forecast; PAUSE IRRIGATION if rainfall > 5 mm | [5]-[8] |
| M4 Fertilizer advisor | Soil test values, crop, area | ICAR nutrient deficit -> Urea/DAP/MOP/SSP bags (50 kg), zinc (25 kg); lime if pH < 6.0, gypsum if pH > 8.0 | [9]-[16] |

| Module | Input | Method | Related work |
|---|---|---|---|
| M5 Profit / ROI | Seven production costs, expected yield and price | Net profit, ROI, break-even price, financial health class | [17] |
| M6 Crop calendar | Crop and sowing date | Six-stage phenological timeline | [7] |
| M7 Price prediction | Historical modal prices | Moving average plus linear momentum for short-term trend | [18]-[22] |
| M8 Mandi locator | User GPS, radius 10-100 km | Haversine distance, live AGMARKNET rates, fallback database | [23]-[26] |
| M9 Administration | Admin account | User audit, scheme/advisory publishing, manual mandi price seeding | [14] |

*Table 4. Modules, methods and the studies that motivate them*

### 4.5.1 Crop recommendation (M2)

The engine evaluates eight soil and climate parameters (N, P, K, pH, rainfall, humidity, temperature and related soil context) against agronomic suitability ranges for each supported crop and returns the best-matching crop, storing inputs and output in crop_recommendations. Unlike learned models [1]-[4], every recommendation can be traced to explicit ranges.

### 4.5.2 Weather and smart irrigation (M3)

The browser supplies GPS coordinates, or a city name is resolved by chaining the Open-Meteo Geocoding API before the Forecast API. The service reads forecast rainfall and evapotranspiration and returns an irrigation advisory. The rule is explicit: if forecast rainfall exceeds 5 mm the system outputs PAUSE IRRIGATION. Requests time out after 5,000 ms and failures return a not-found error rather than substituted data.

### 4.5.3 Fertilizer advisor (M4)

The advisor compares soil-test nutrient levels with ICAR-standard requirements to obtain elemental deficits for nitrogen, phosphorus and potassium, then converts them into commercial products: Urea, DAP, MOP and SSP in 50 kg bags and zinc in 25 kg bags, using rounding rules to give whole-bag quantities. Soil amendments are triggered by strict boundaries: lime when $pH < 6.0$ and gypsum when $pH > 8.0$. Warnings are issued for acidic and alkaline soil so that the system never recommends a chemical load that could cause soil toxicity. This directly targets the actionability gap identified in [14].

### 4.5.4 Profit and ROI engine (M5)

The engine aggregates seven production cost heads and compares them with projected gross revenue. Using the cost-concept approach of [17]:

$$Net\ profit = Gross\ revenue - Total\ cost$$

$$ROI\ (\%) = (Net\ profit / Total\ cost) \times 100$$

$$Break\text{-}even\ price = Total\ cost / Expected\ yield$$

A financial-health classifier labels the outcome from these values.

### 4.5.5 Crop calendar and price prediction (M6, M7)

The crop calendar produces a six-stage phenological timeline for the selected crop. The price engine projects short-term APMC commodity prices using moving averages and linear momentum computed on stored market_prices history. This is deliberately simpler than the ARIMA and machine-learning models in [18]-[21]; it is transparent and cheap, and is identified as the first upgrade target in Section 7.

### 4.5.6 Nearby mandi locator (M8)

Given the user's coordinates and a radius between 10 and 100 km, the great-circle distance to each registered market is computed with the Haversine formula:

$$d = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{lat2 - lat1}{2}\right) + \cos(lat1)\cos(lat2)\sin^2\left(\frac{lon2 - lon1}{2}\right)}\right), \quad R = 6371 \text{ km}$$

Markets inside the radius are returned with live prices from the data.gov.in AGMARKNET API and a Google Maps navigation link. Because the API is intermittently unavailable, the service falls back to a verified database and labels each result with its data source (live or fallback/cached) so the user is never misled.

### 4.5.7 Administration (M9)

Administrators audit farmers, publish government schemes and advisories, and manually seed mandi prices, giving a governed source for the fallback dataset.

## 5. Implementation

The backend is a Java 17 Spring Boot 3.3.4 application organised into config (SecurityConfig, JwtTokenFilter), controller (AuthController, FarmerController, MandiController, among others), dto, model (JPA entities such as User, Farm and MarketPrice), repository and service packages. The frontend is a static site in HTML5, CSS3, ES6+ and Bootstrap 5.3 with an Agro Forest Green theme (#2e7d32), Inter typography and Bootstrap Icons; api.js centralises Fetch calls and JWT injection. Deployment is containerised with Docker for the Spring Boot executable jar, targeting Netlify for the frontend and Render for the backend. The code is available at https://github.com/imyash058-hub/krishi-management-system.

## 6. Testing and Results

The team executed integration and end-to-end tests with JUnit 5 and MockMvc. Sixteen core test cases ran with 0 failures and 0 errors. Table 5 summarises the focus areas.

| Area | Verified behaviour |
|---|---|
| Security integration | Unauthenticated access to /api/farmers/ and /api/admin/ routes rejected |
| Weather latency and fallback | Open-Meteo calls time out after 5,000 ms and return a 404 ResourceNotFoundException |
| Agronomic boundaries | testFertilizerAcidicAndAlkalineSoilWarnings: no unsafe chemical load recommended for acidic or alkaline soil |
| Data failover | testNearbyMandiPricesJaipur: UI shows the cached-data tag when the live government API fails |

*Table 5. Test coverage summary (16 test cases, 0 failures, 0 errors)*

These tests verify correctness of security rules, failure handling and agronomic safety limits. They do not measure agronomic outcomes: the platform has not yet been evaluated in a field trial with farmers, so claims about yield or income improvement are not made. Likewise, the 250 ms and 1.5 s figures are design targets stated in the requirements; measured latency under load should be reported once benchmarked.

## 7. Discussion, Limitations and Future Work

Smart Krishi addresses each problem identified in Section 3 with a transparent rule or calculation, and its main distinguishing features are integration, actionable outputs and honest data handling, which responds to the forecast-credibility concern in [8]. Because outputs are rule-based, they are easy to audit and explain to a mentor or an agronomist, but they cannot learn from data. Three limitations are important. First, the price engine is simpler than published time-series and machine-learning methods [18]-[21]. Second, evidence from advisory services shows trust and adoption without systematic yield gains [27], so benefits should be treated as decision support until field-tested. Third, e-NAM and market studies show mixed price outcomes [24], [25], so the mandi locator is an information tool rather than a guarantee of better prices.

Future work includes ARIMA or hybrid forecasting validated with MAPE and RMSE on AGMARKNET history, a hybrid crop recommender that keeps rules for safety and uses learned ranking, regional language and voice interfaces for low-literacy users, a field pilot with farmers measuring adoption and input savings, and measured performance benchmarking against the stated latency targets.

## 8. Conclusion

This paper reviewed 32 studies and showed that soil-aware fertilizer advice, forecast-based irrigation, price information and market access each have documented benefits, while adoption barriers call for simple, trustworthy and low-cost tools. Smart Krishi combines these decisions into one verified, rule-based, mobile-friendly platform with secure role-based access and a failover design that never fabricates data. Sixteen integration tests passed, and the next step is to validate the platform with real farmers and stronger forecasting models.

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